

MTH202 MID TERM BY IQRA AZHAR

MTH202

Mid term

2023

Solved by

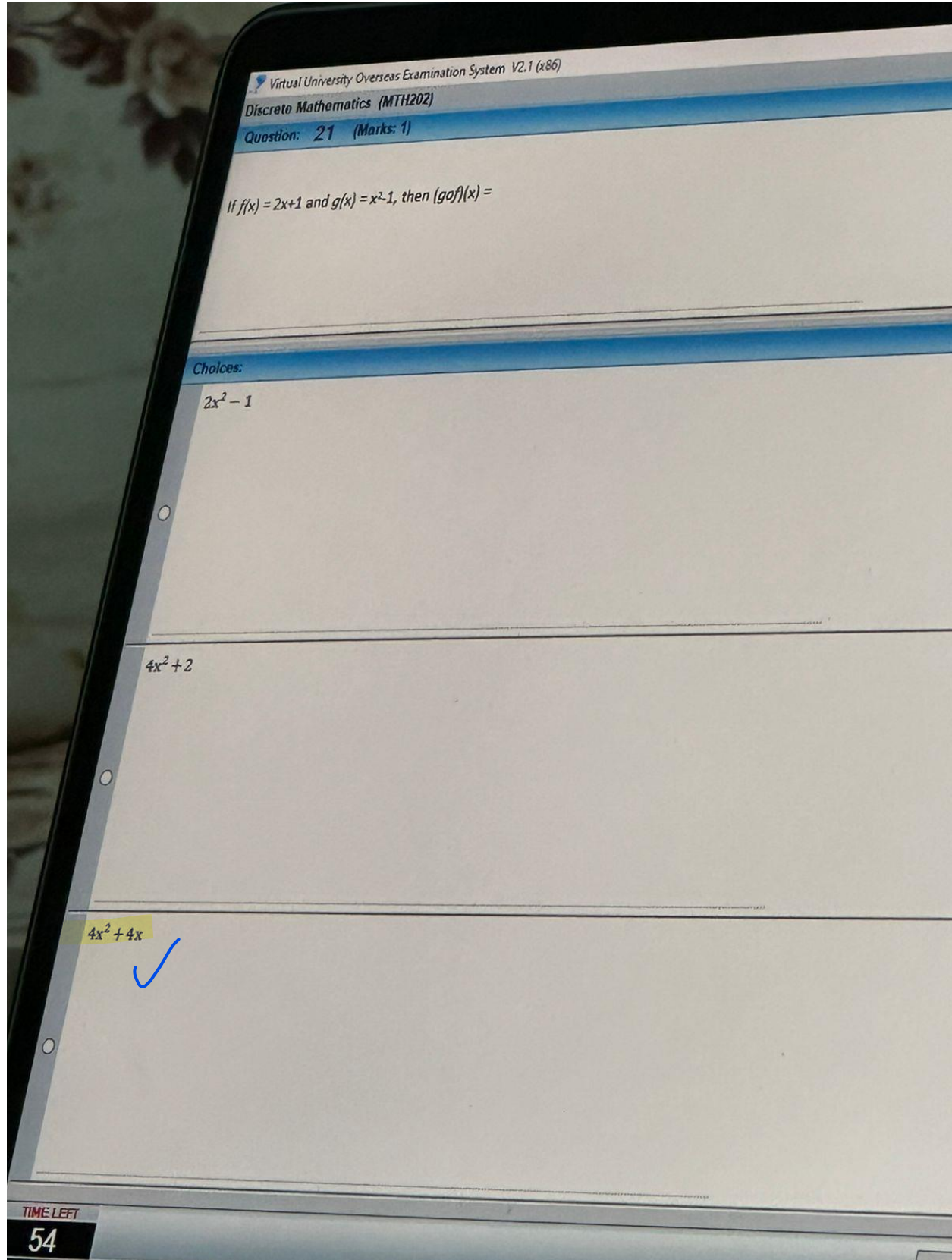
 Iqra Azhar 

YTUBE

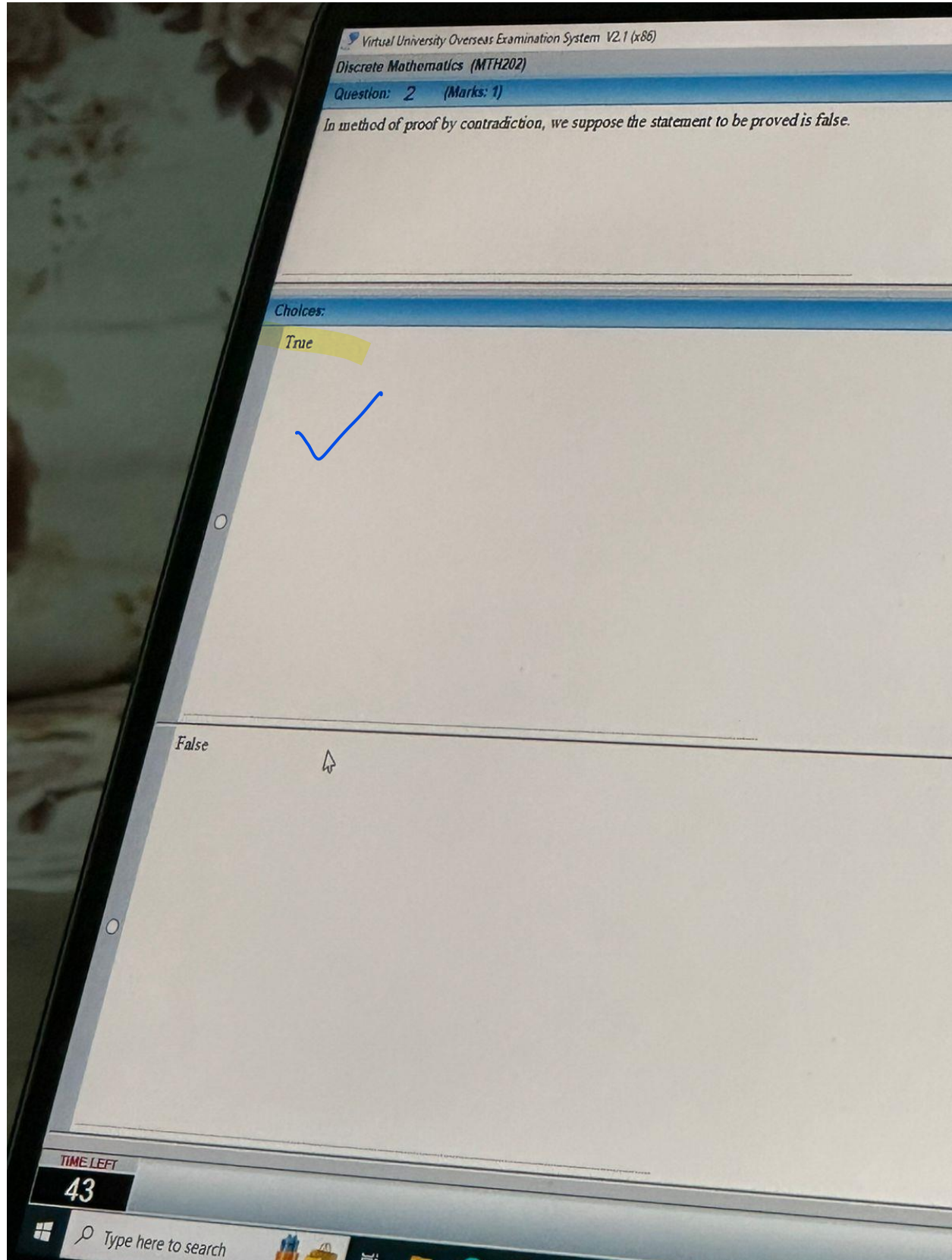
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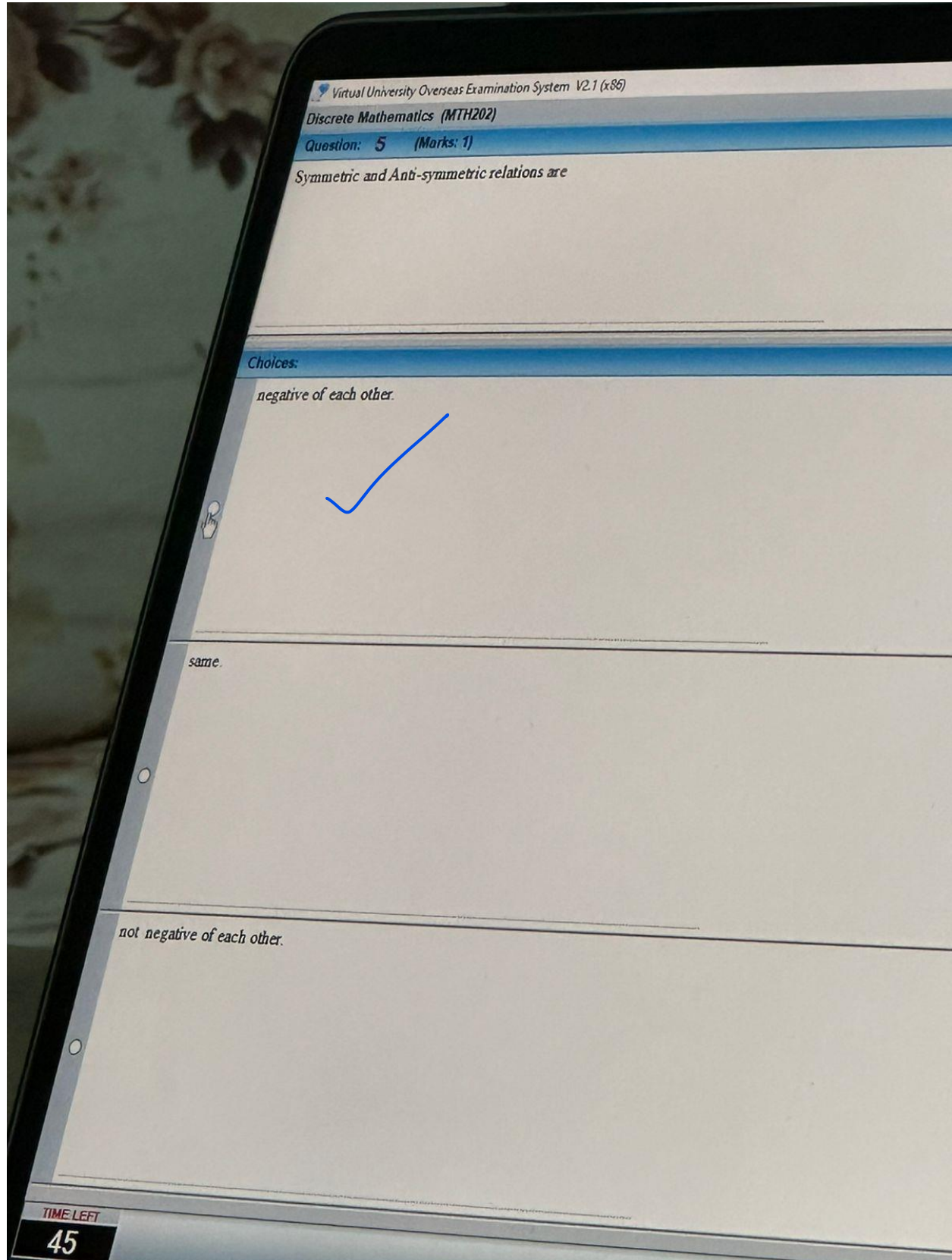
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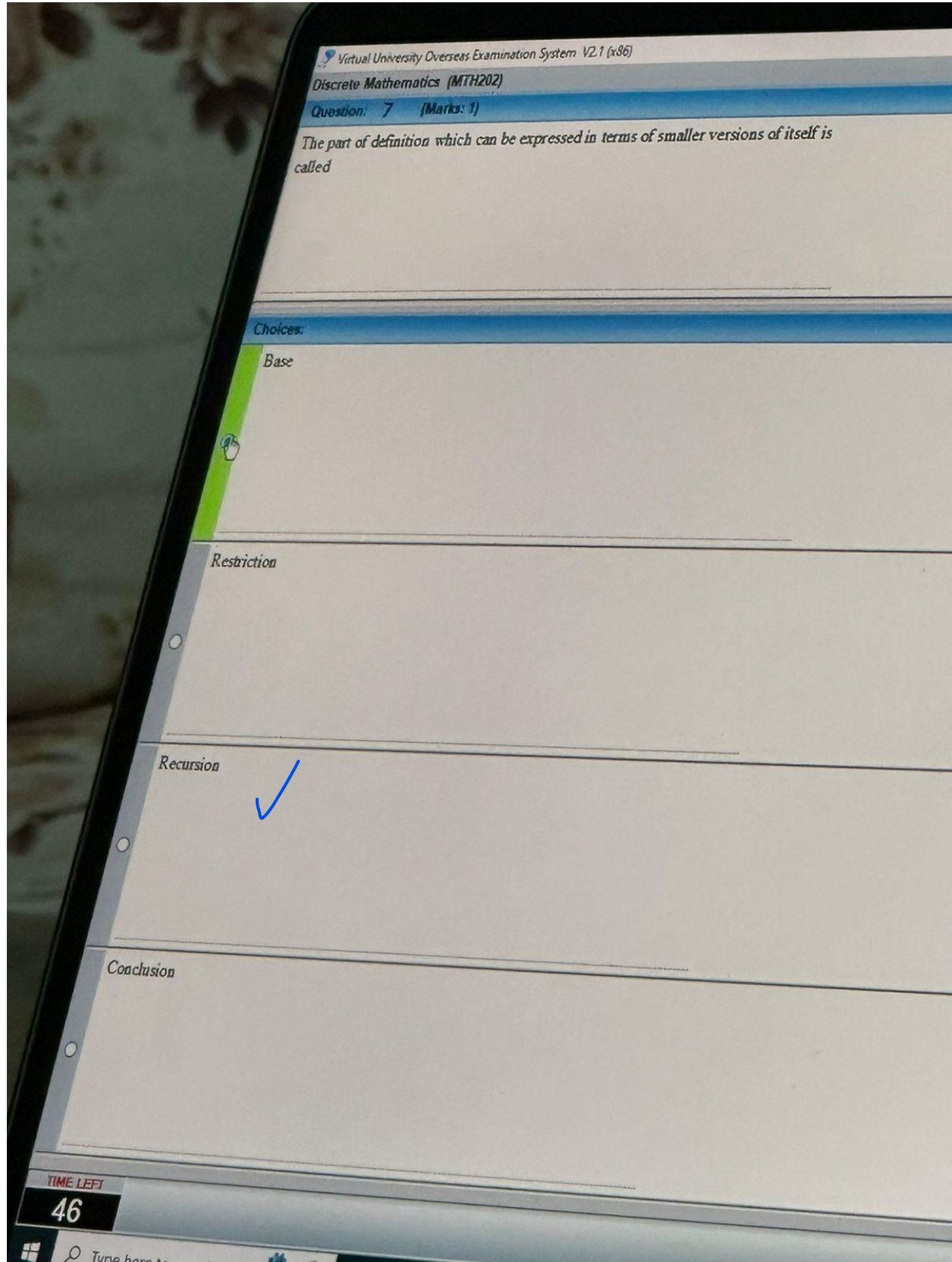
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Virtual University Overseas Examination System V2.1 (x86)

Discrete Mathematics (MTH202)

Question: 10 (Marks: 1)

Subset of $A \times B$ is called a

Choices:

☐ Relation on A

☐ Relation on B

☒ Relation from A to B ✓

☐ Relation from B to A

TIME LEFT
47

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Virtual University Overseas Examination System V2.1 (x86)

Discrete Mathematics (MTH202)

Question: 12 (Marks: 1)

Sixth term of the sequence $a_n = 2n + 9$ is

Choices:

25
☐

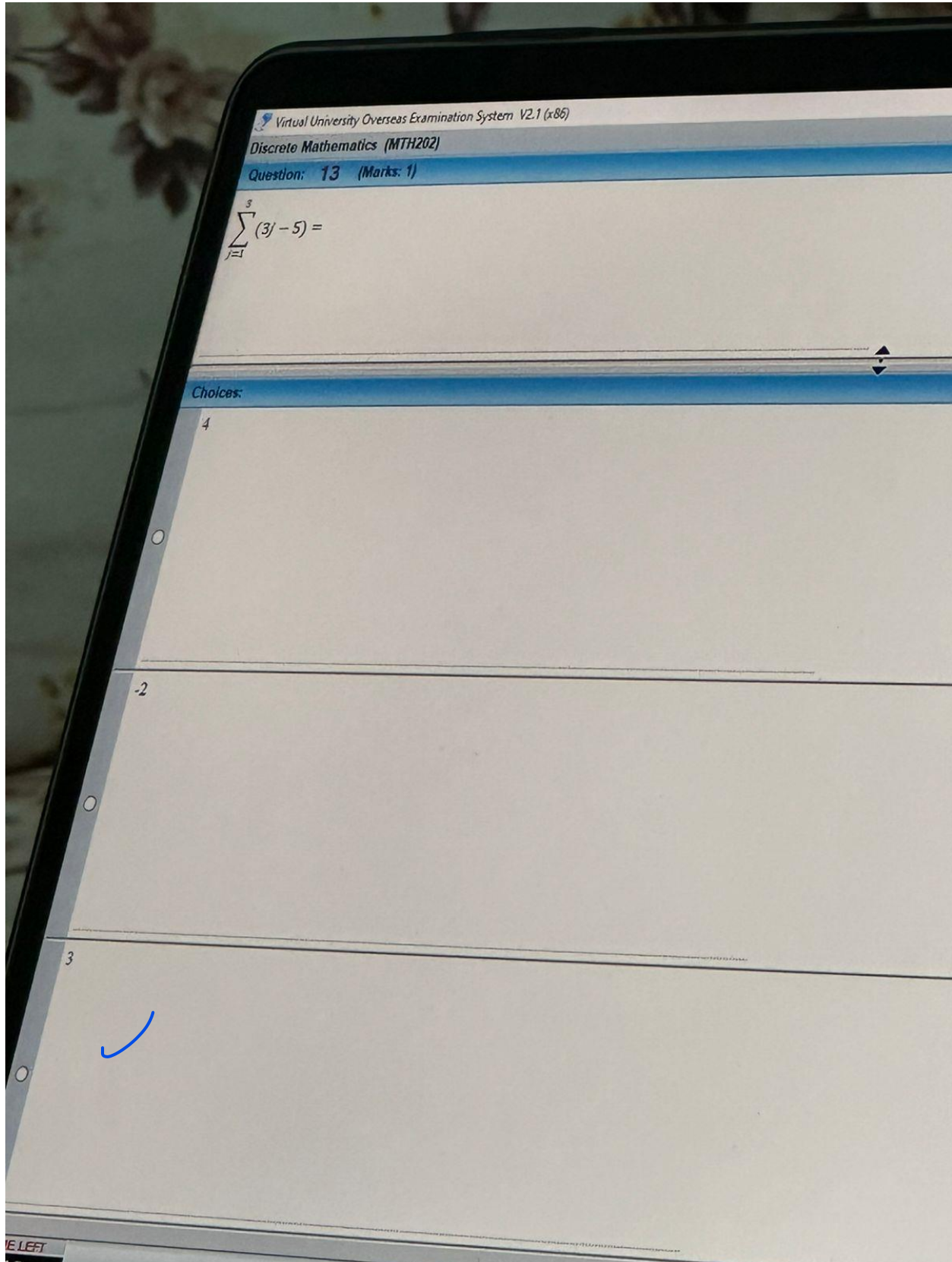
27
☐

23
☐

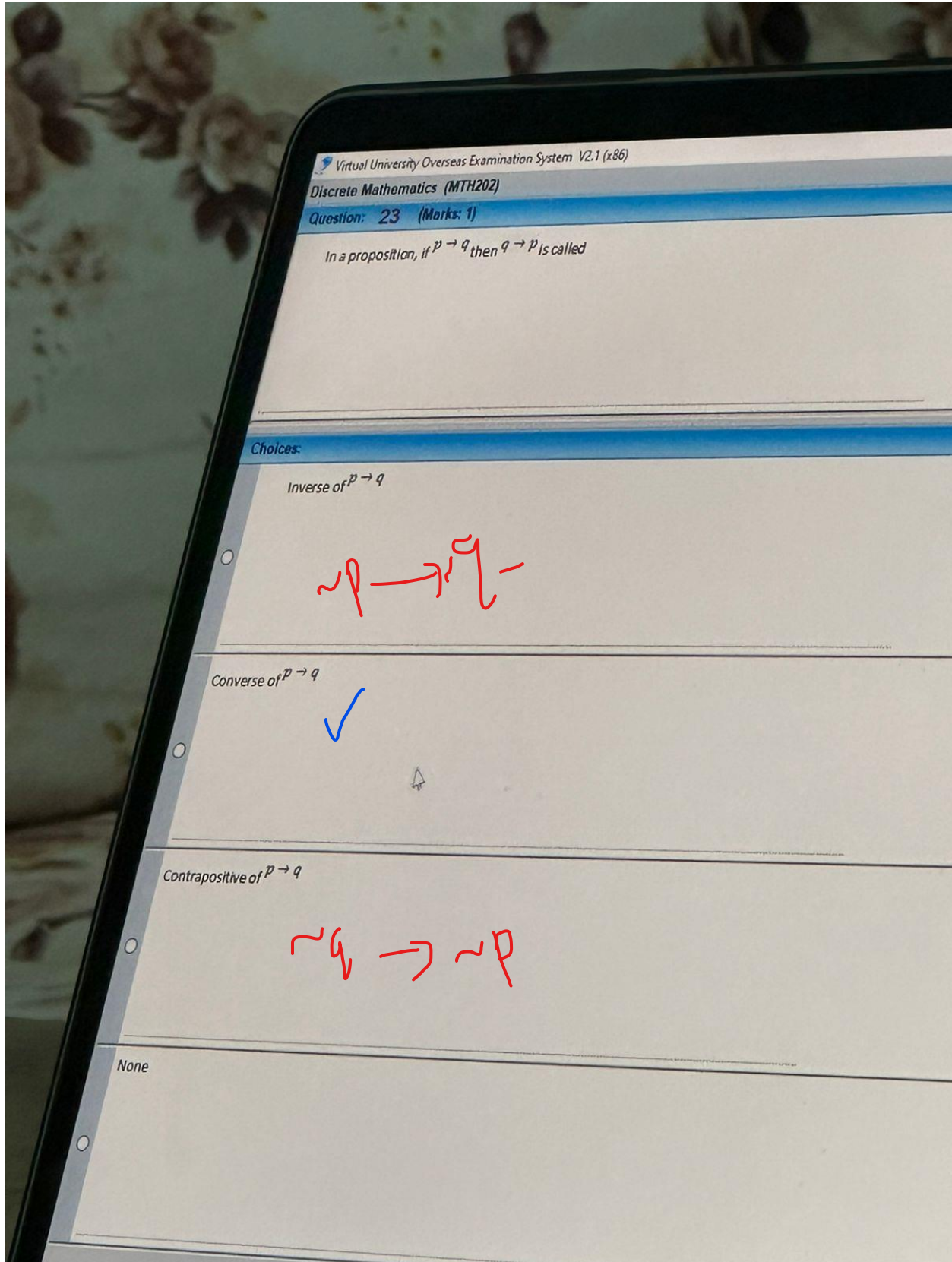
21
☒

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49

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Virtual University Overseas Examination System V2.1 (x86)

Discrete Mathematics (MTH202)

Question: 24 (Marks: 1)

If P and Q are two disjoint sets then $P \cap Q$ is _____.

Choices:

☒ $\emptyset$

☐ P

☐ $P \cup Q$

☐ Q

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55

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Question: 25 (Marks: 1)

The Relation $R = \{(1,2), (1,3), (2,3), (3,1)\}$ on the set $A = \{1,2,3\}$. Then the matrix Representation of R is

Choices:

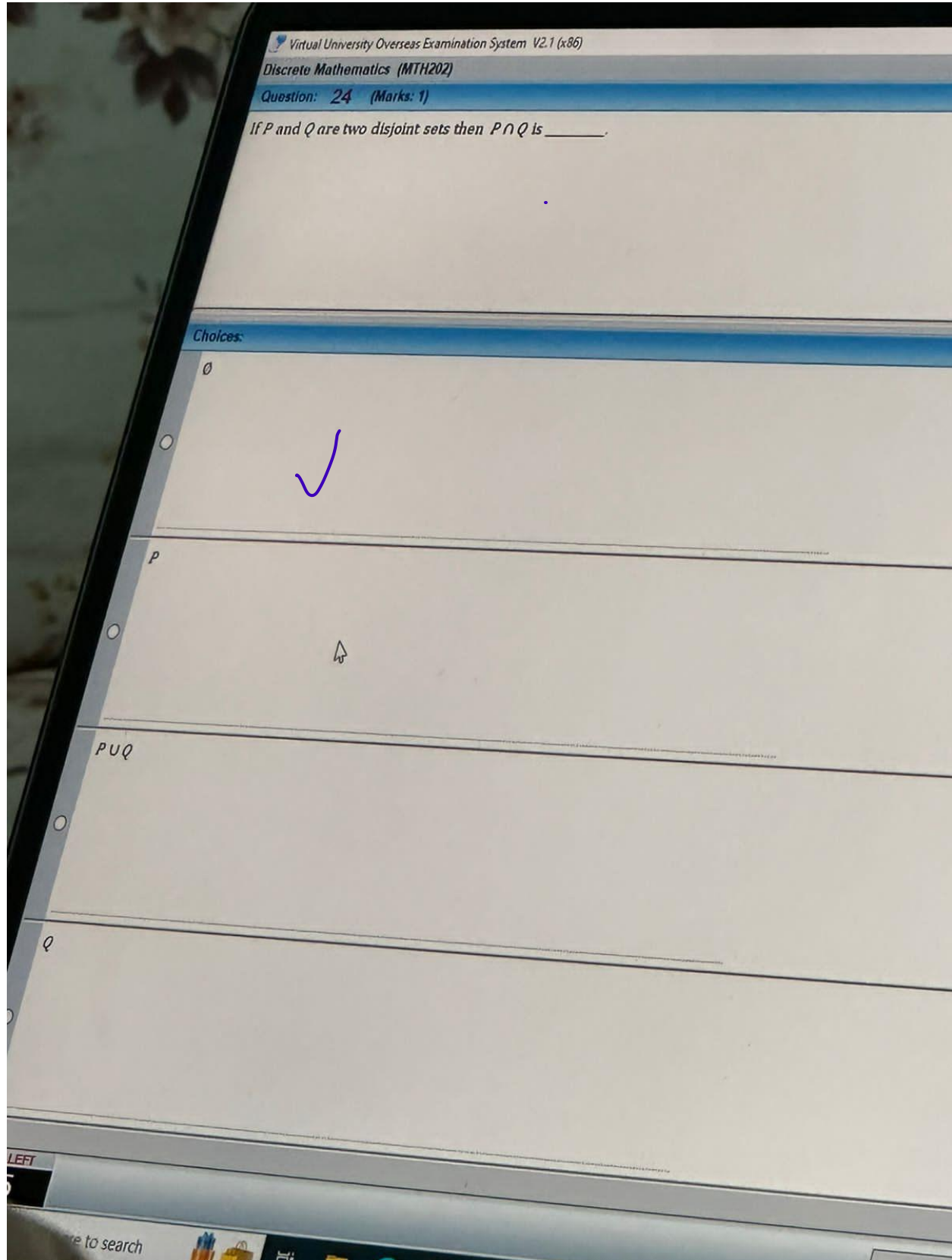
☐ $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

☐ $\begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

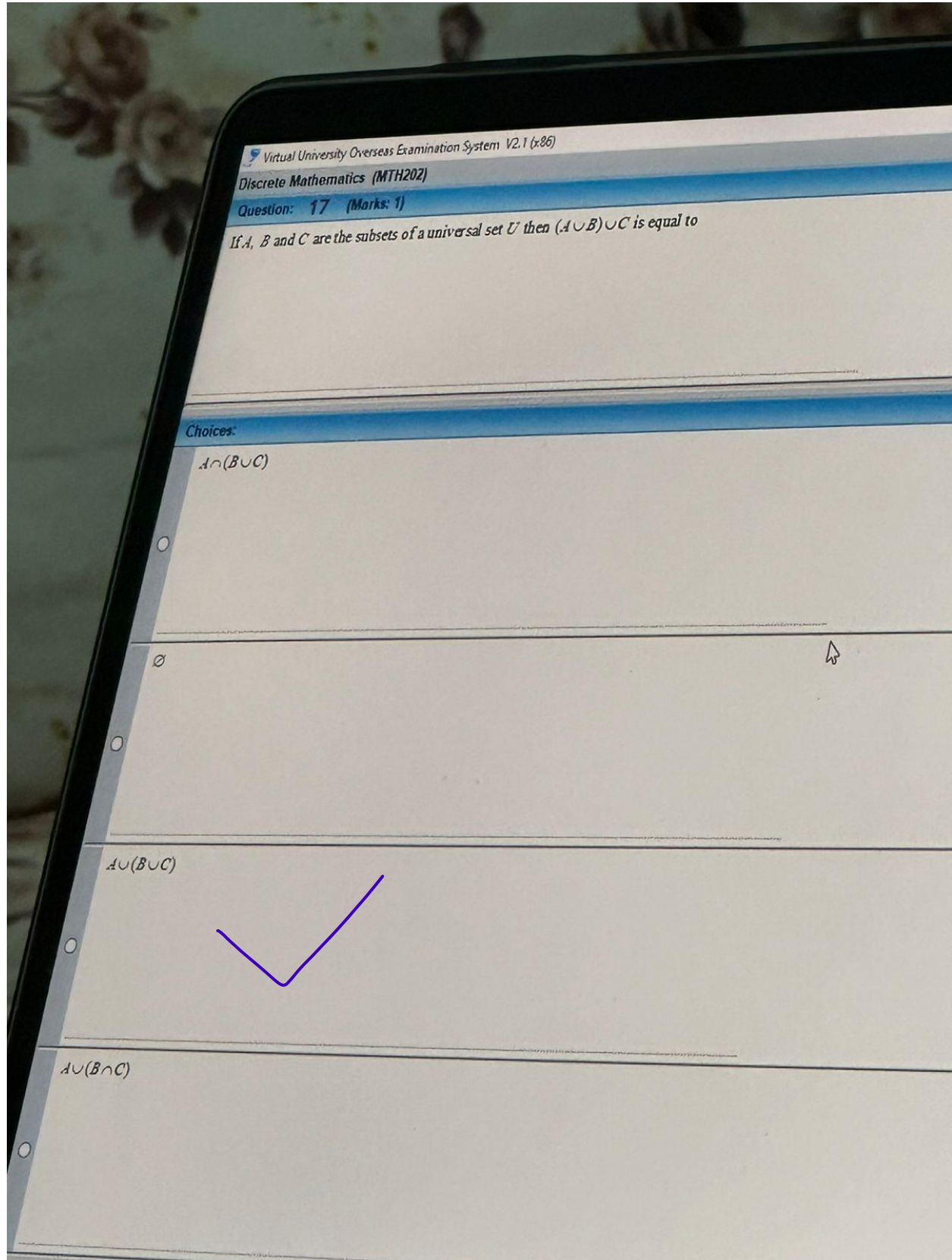
☒ $\begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ ✓

☐ $\begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$

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Discrete Mathematics (MTH202)

Question: 19 (Marks: 1)

Number of terms in the expansion of $(1+x)^6$.

Choices:

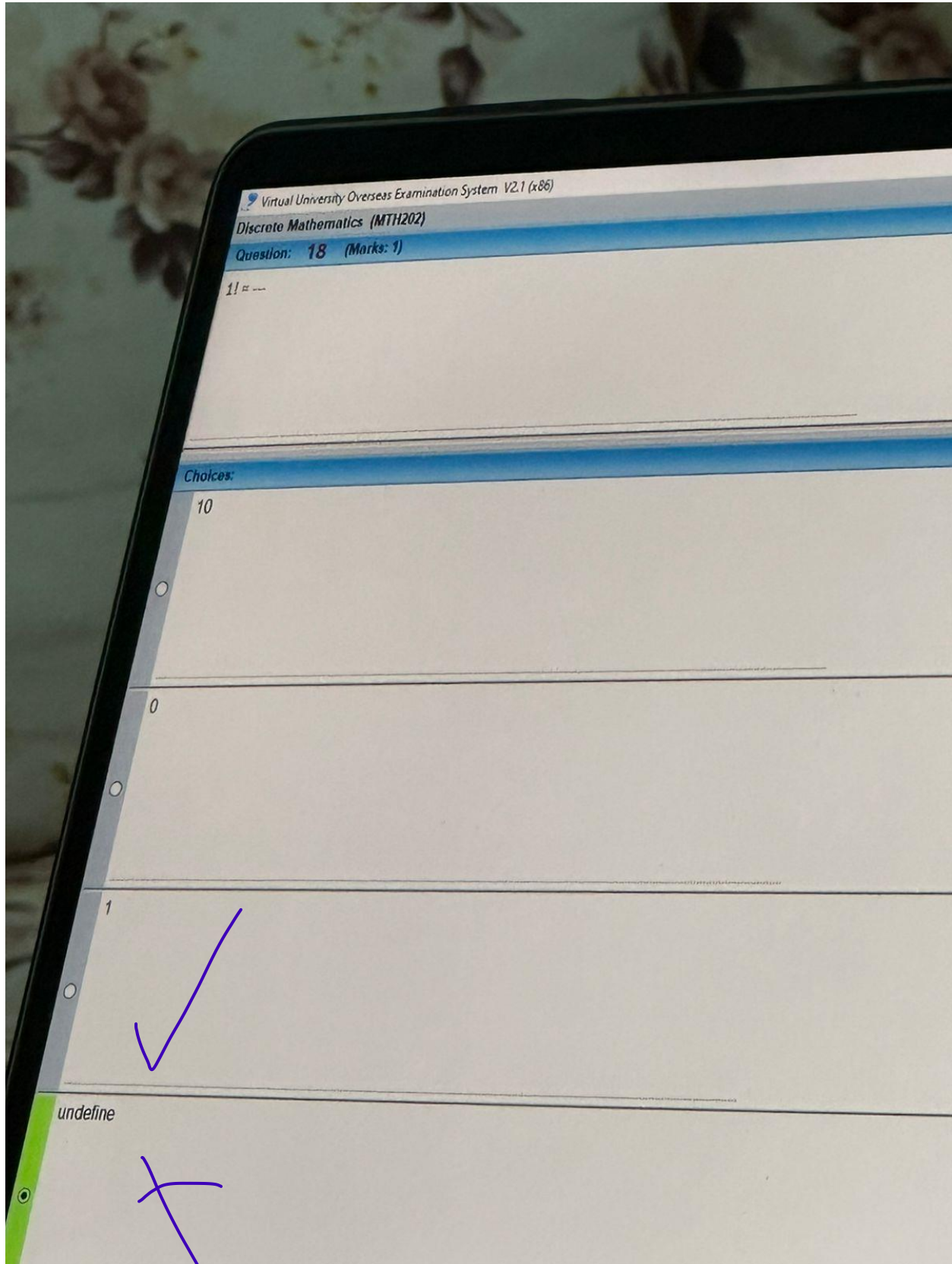
☐ 5

☒ 7

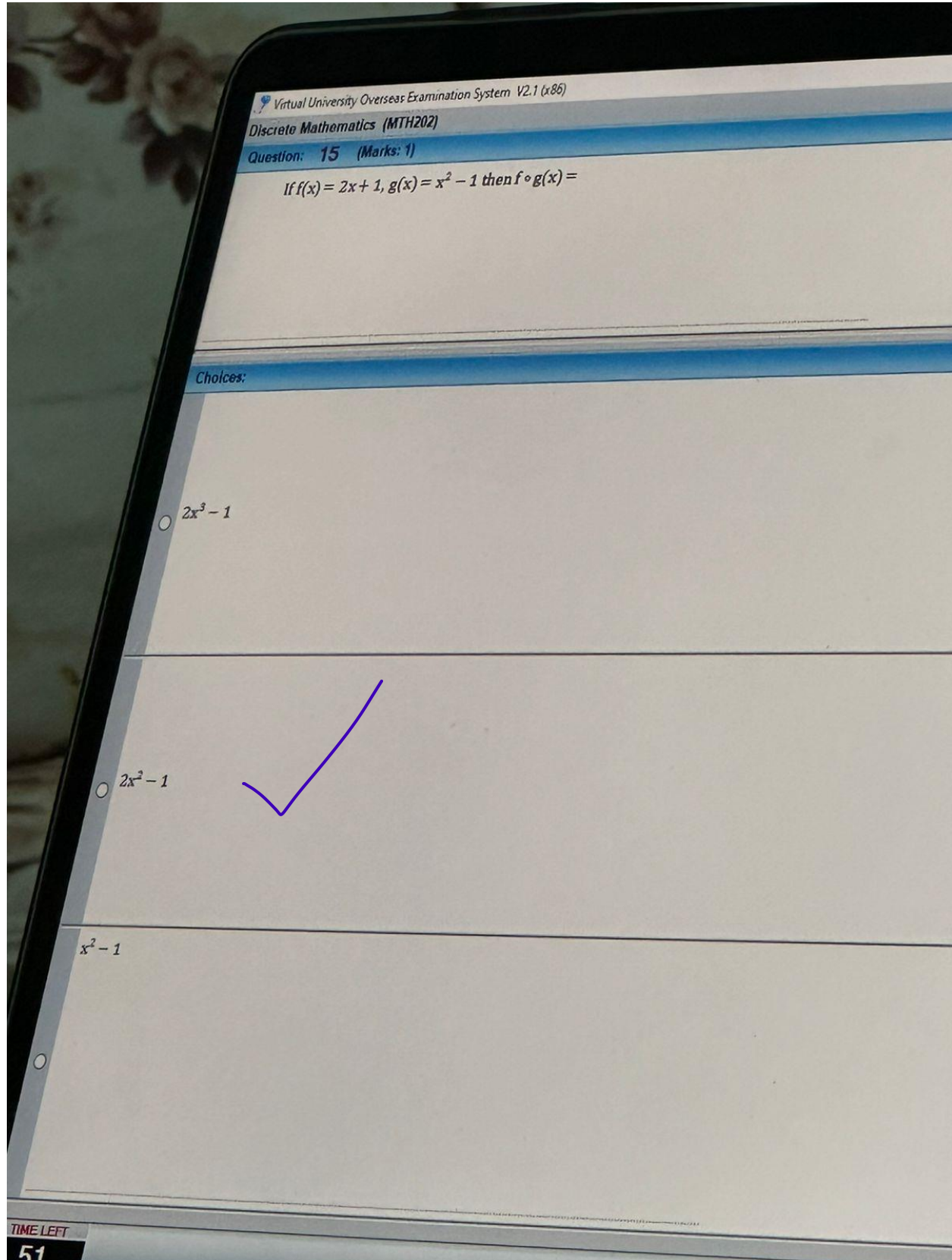
☒ 6

☐ none

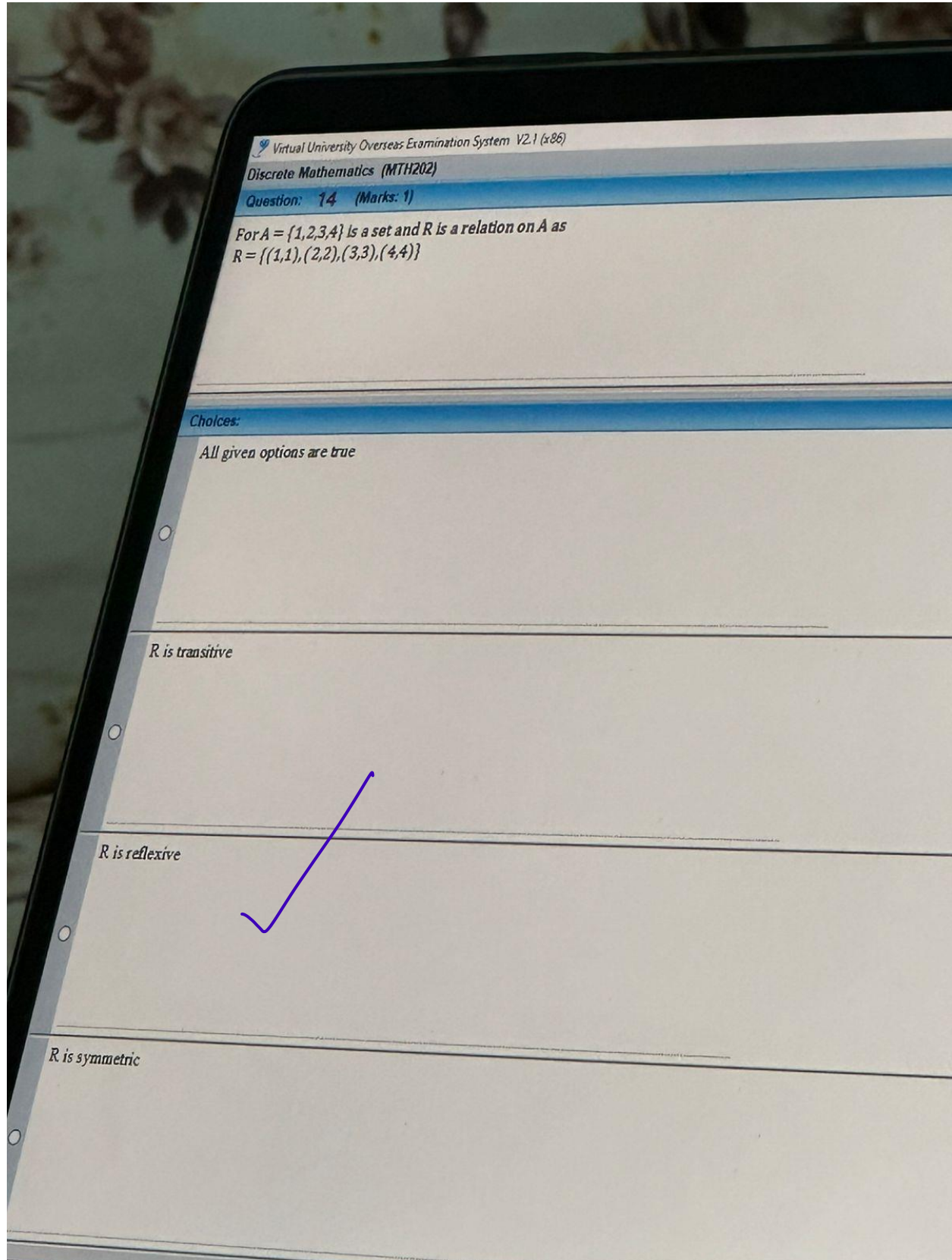
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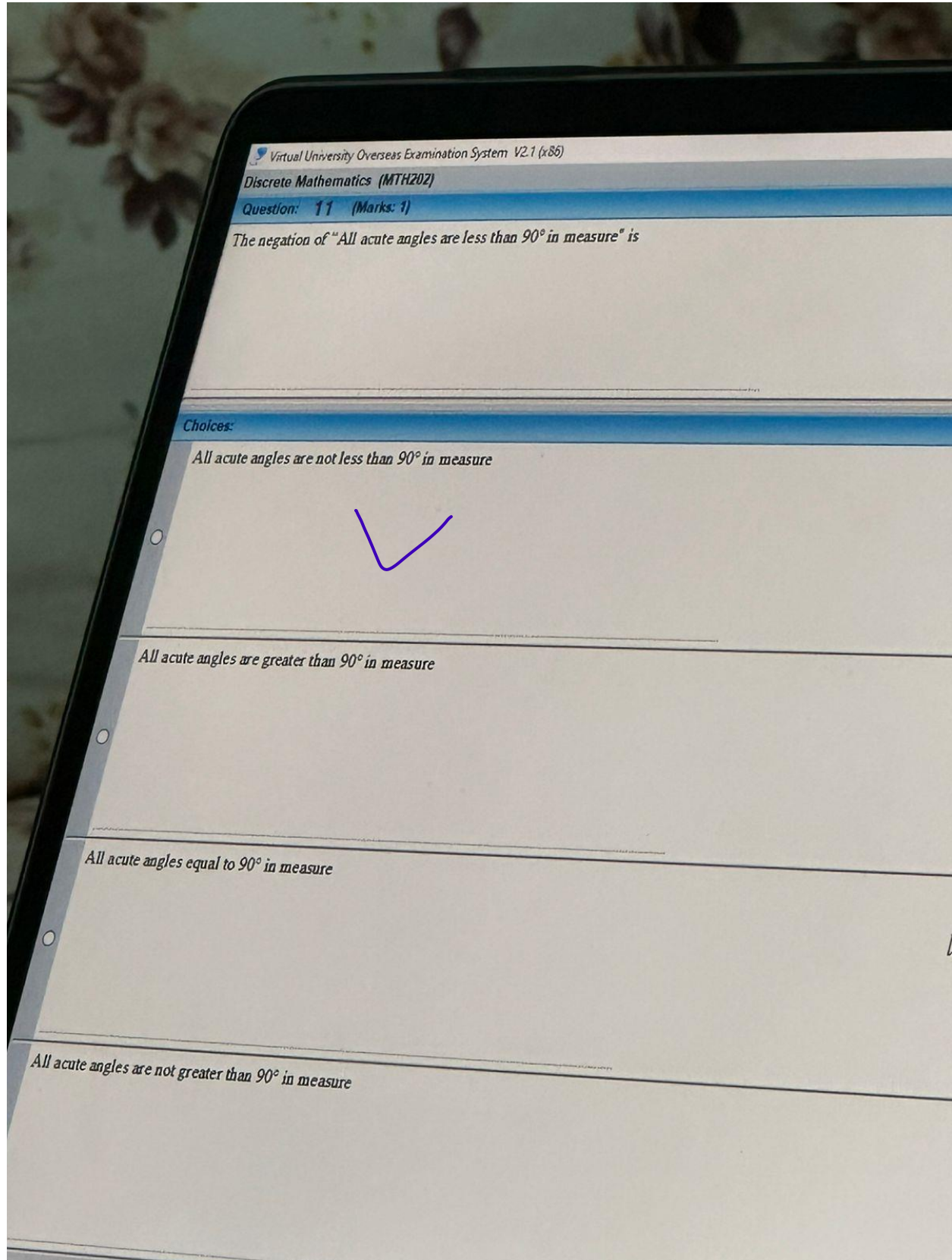
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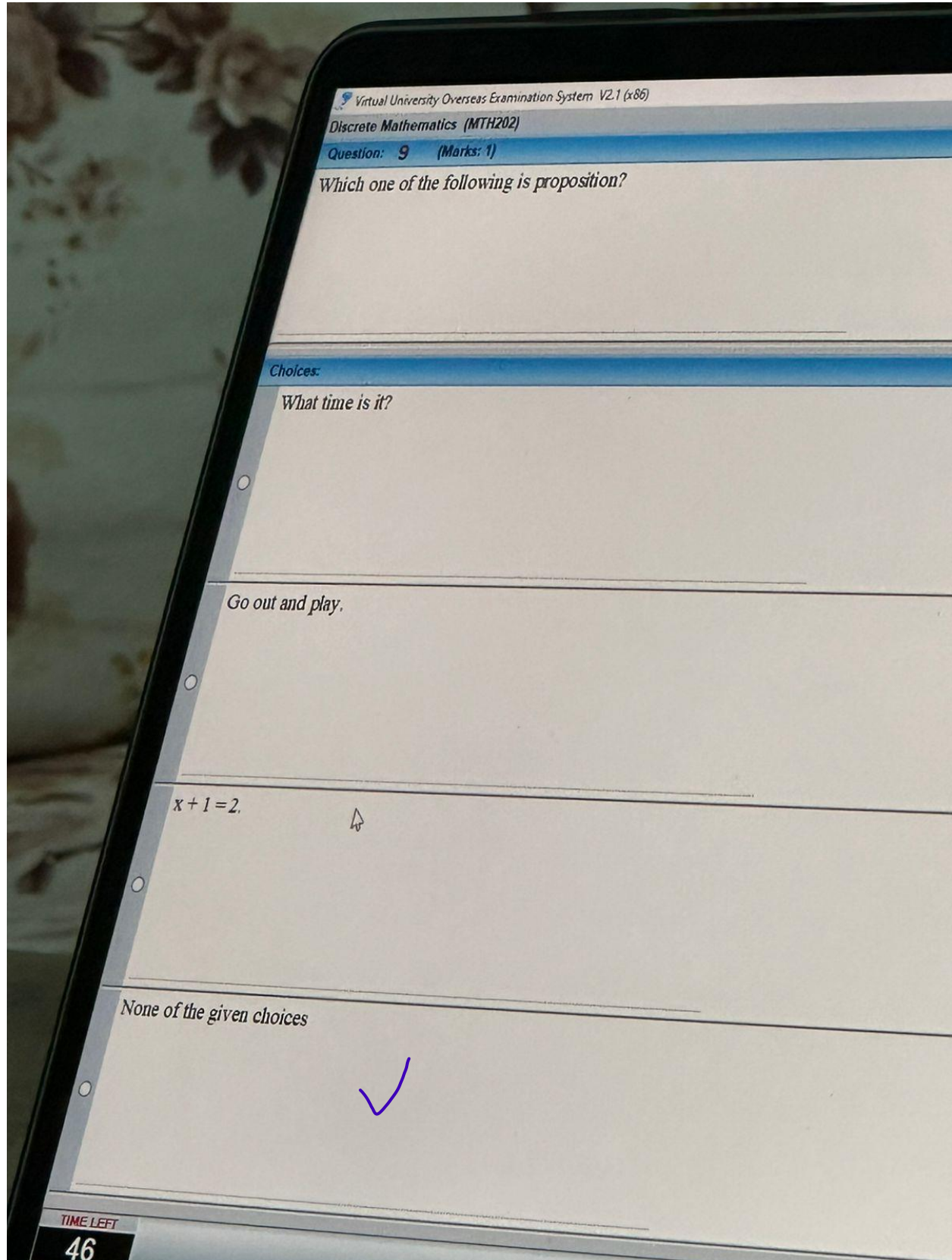
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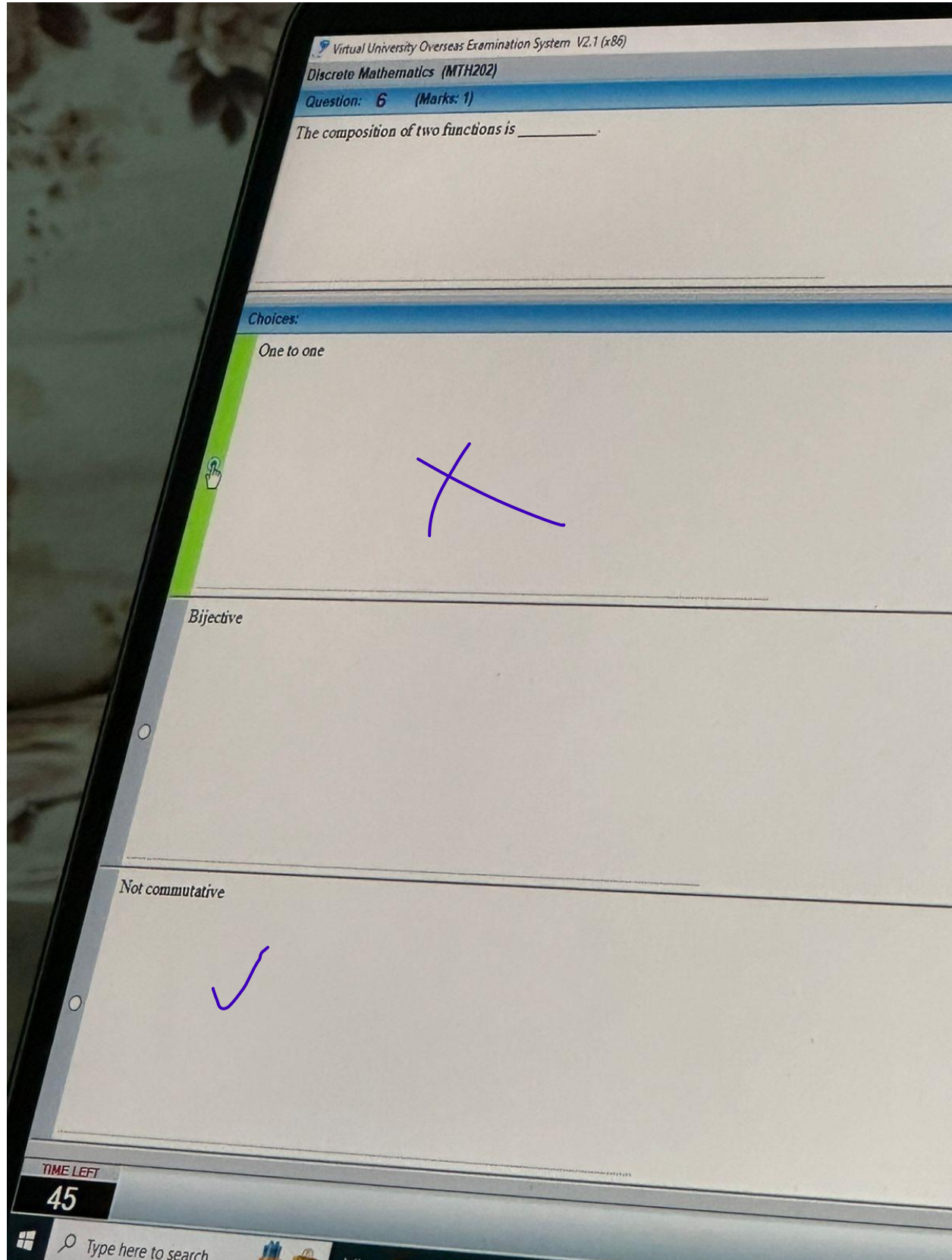
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Virtual University Overseas Examination System V2.1 (x86)

Discrete Mathematics (MTH202)

Question: 1 (Marks: 1)

The following directed graph can be represented in form of matrix as



Choices:

☐ $\begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$

☐ $\begin{pmatrix} 2 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 2 \end{pmatrix}$

☐ $\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 2 \end{pmatrix}$

☐ $\begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

TIME LEFT
43

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Virtual University Overseas Examination System V2.1 (x86)

Discrete Mathematics (MTH202)

Question: 28 (Marks: 3)

A man deposited Rs. 10 in a bank in the first month; Rs. 15 in the second month; Rs. 20 in the third month and so on. Find how much he will have deposited in the bank by the 9th month.

$d = 15 - 10 = 5$ $a = 10$ $n = 9$

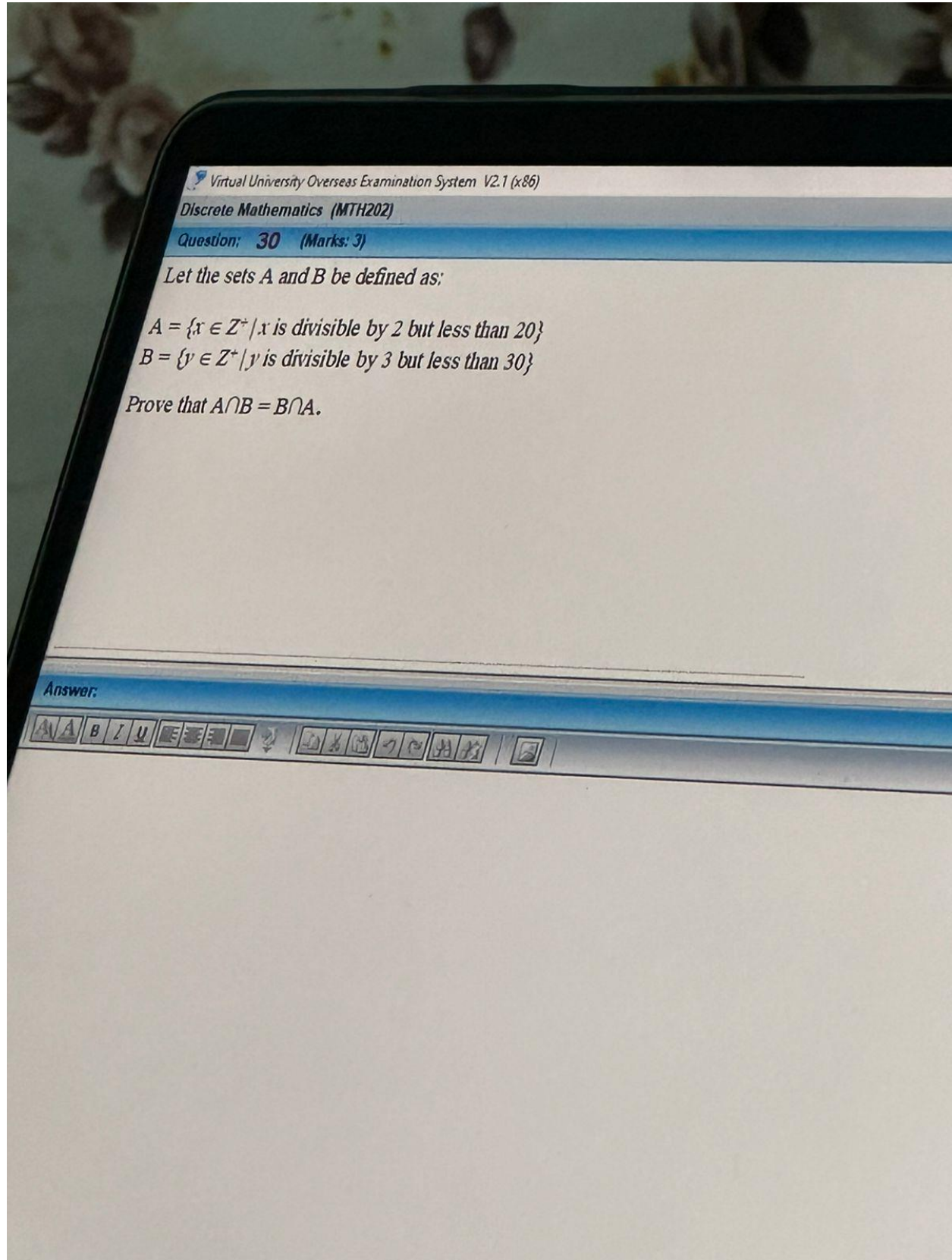
$a_n = a_1 + (n-1)d$

$a_9 = 10 + (9-1)5$

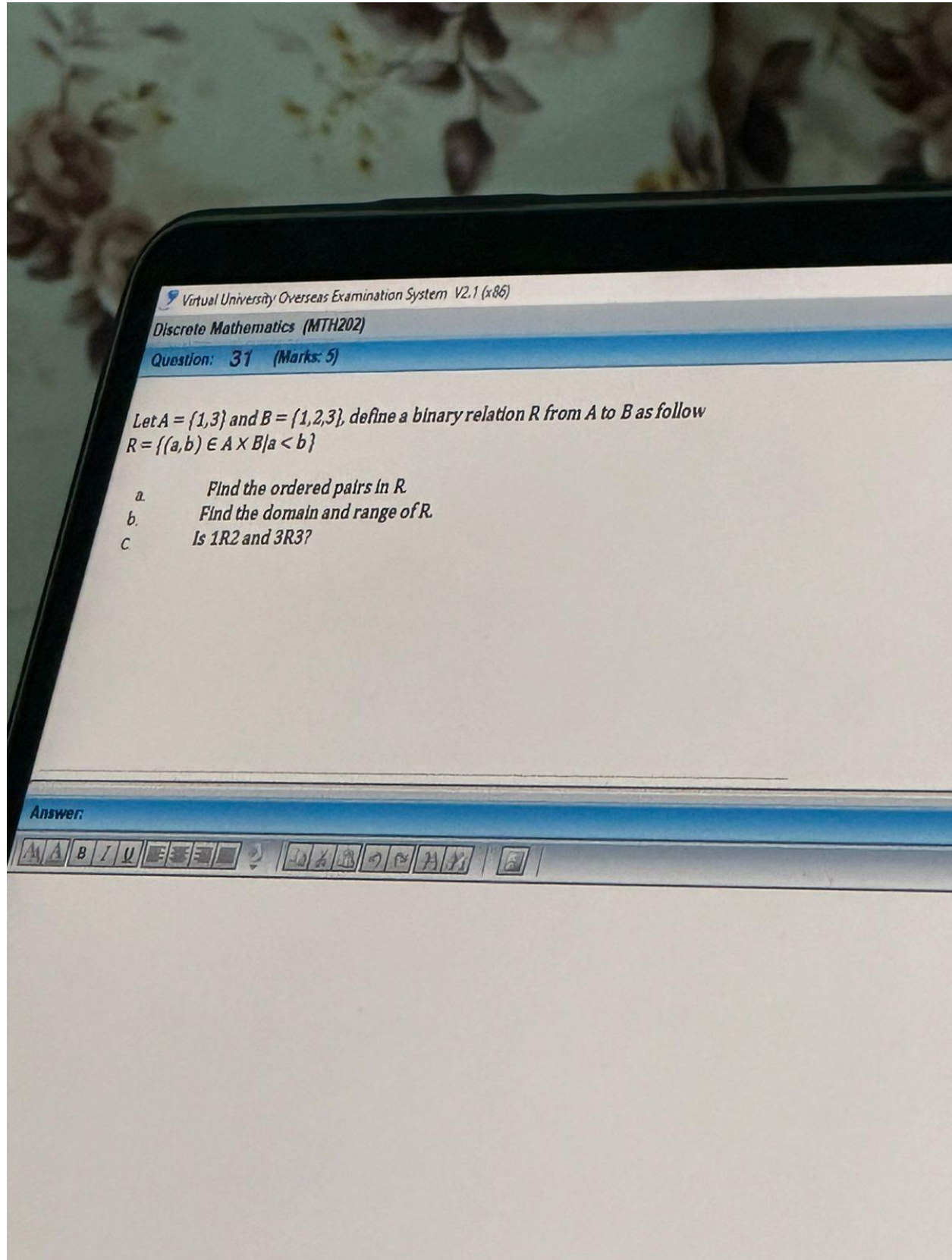
$a_9 = 10 + (8)5$

$a_9 = 10 + 40$
 $= 50$

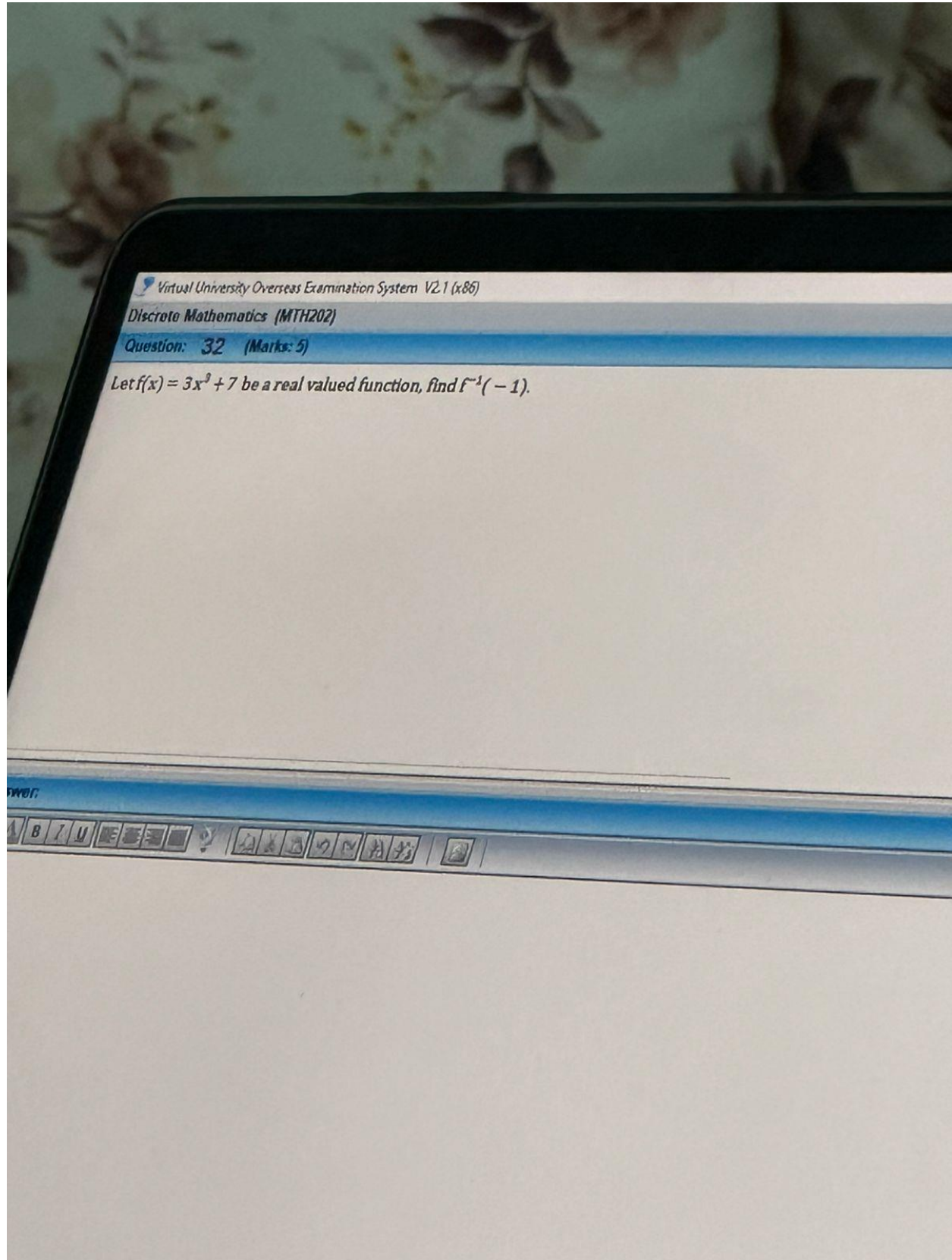
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Virtual University Overseas Examination System V2.1 (x86)

Discrete Mathematics (MTH202)

Question: 33 (Marks: 5)

Determine whether the following is a tautology, a contingency or an absurdity.

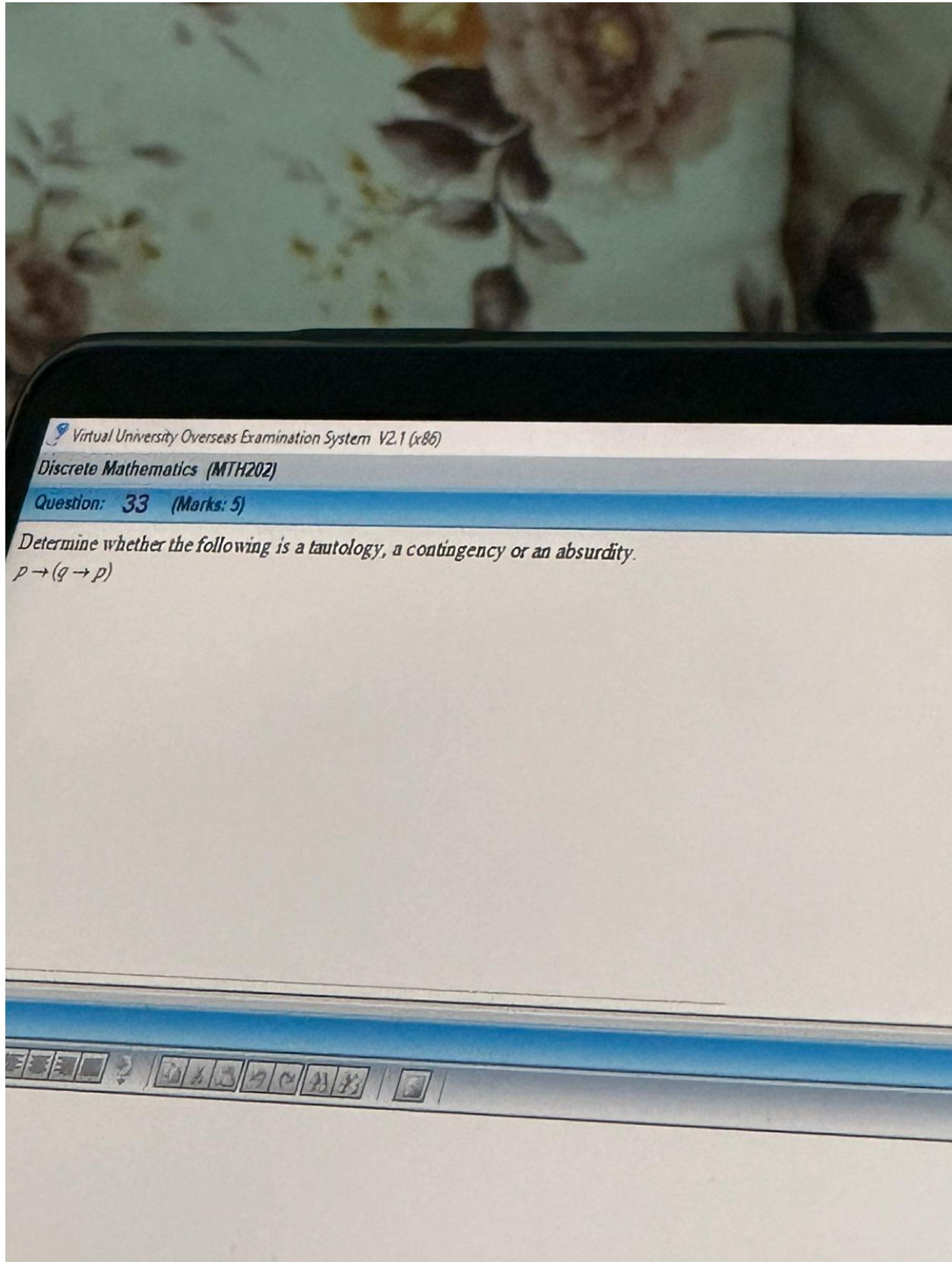
$$p \rightarrow (q \rightarrow p)$$

p	q	$q \rightarrow p$	$p \rightarrow (q \rightarrow p)$
T	T	T	T
T	F	T	T
F	T	F	T
F	F	T	T

Answer:

It is tautology

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Virtual University Overseas Examination System VZ.1 (x86)

Discrete Mathematics (MTH202)

Question: 29 (Marks: 3)

Suppose $A = \{1, 2, 3\}$ and $B = \{x, y, z\}$ are two sets and R is a relation from A to B as $R = \{(1, y), (2, x), (2, z), (3, z)\}$ then determine the matrix representation of R .

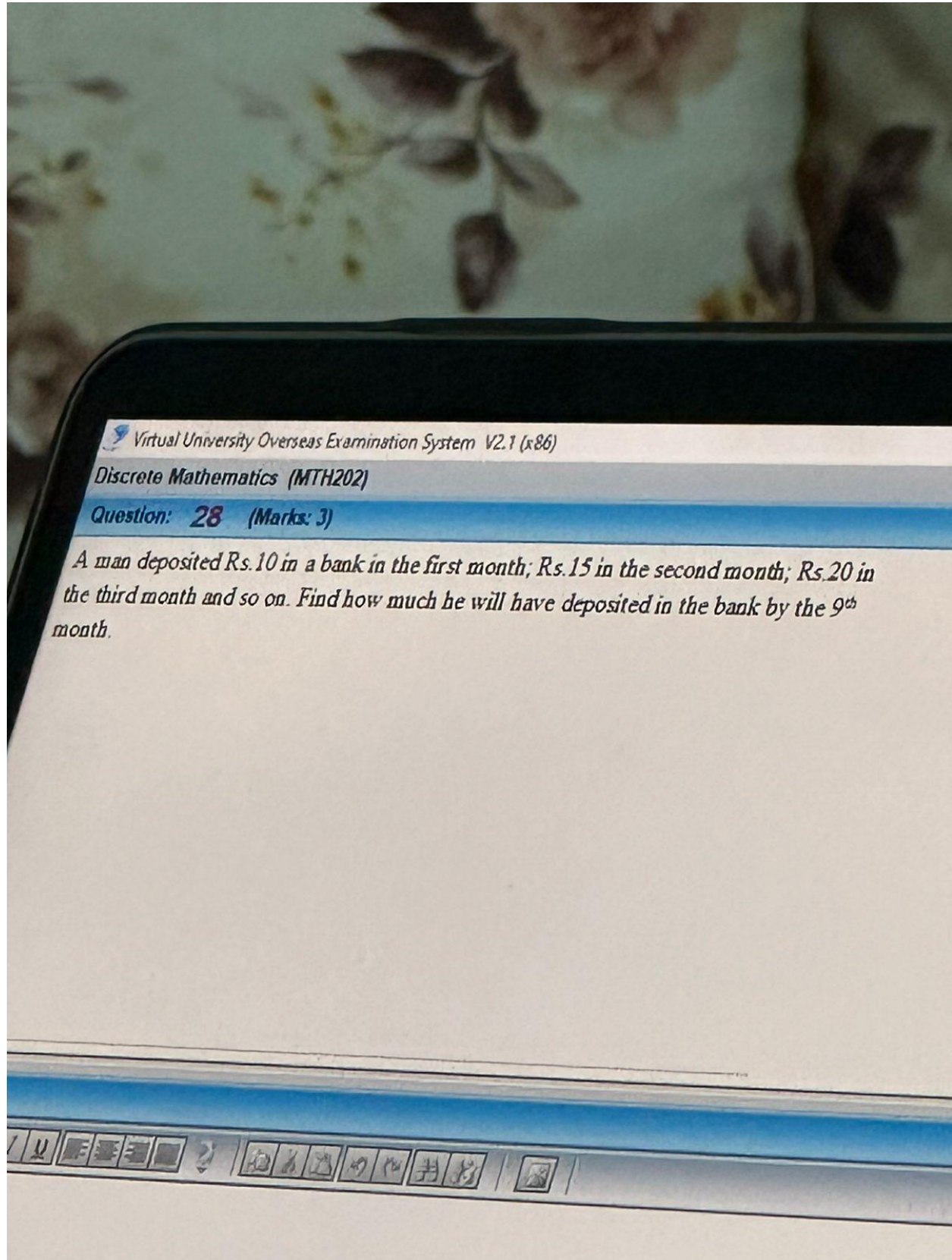
Handwritten solution:

The matrix representation of R is a 3×3 matrix. The rows are indexed by elements of A (1, 2, 3) and the columns are indexed by elements of B (x, y, z). The matrix is:

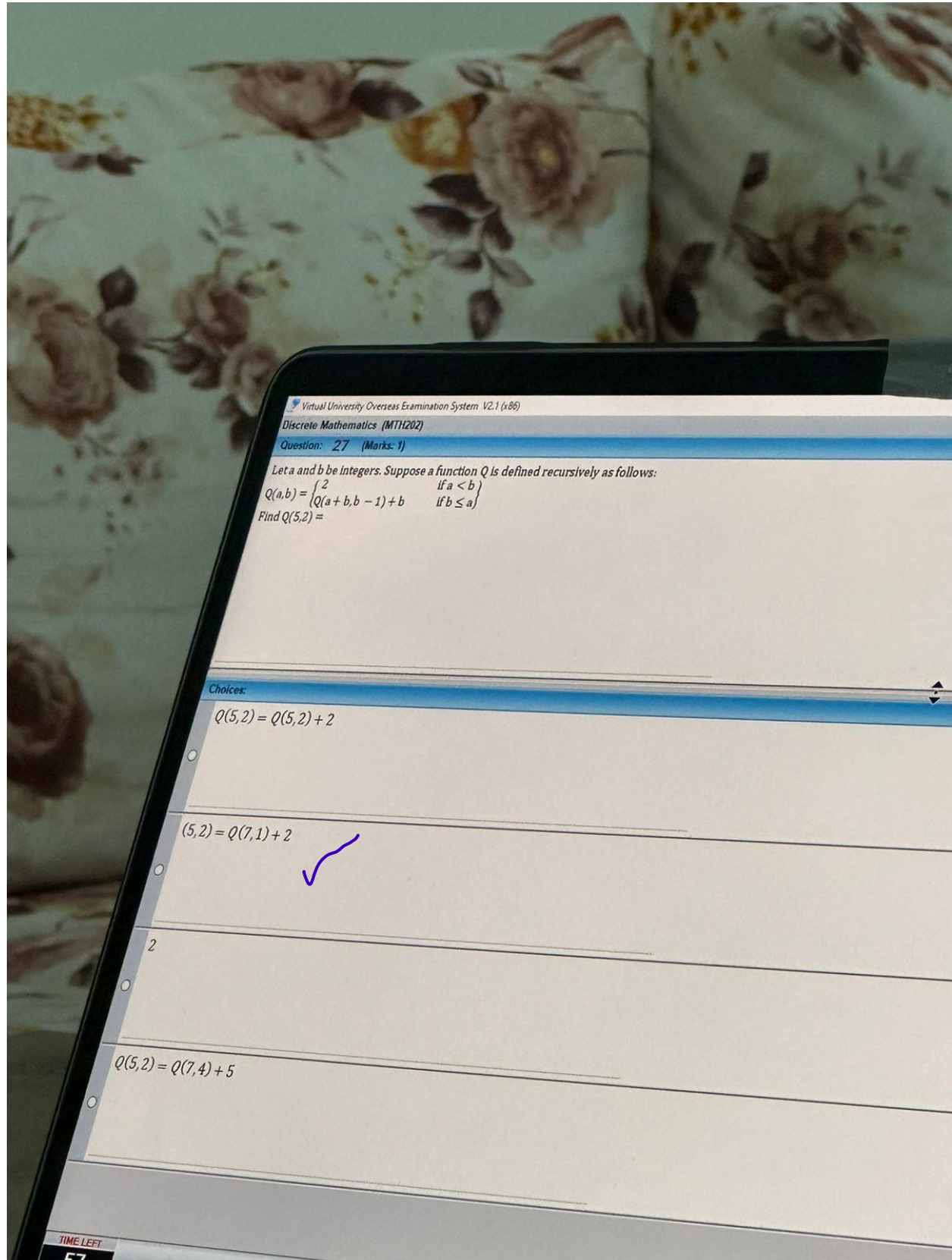
$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

The matrix is shown with handwritten annotations: the first row is labeled with 1, 2, 3; the first column is labeled with x, y, z; and the matrix is enclosed in a large bracket.

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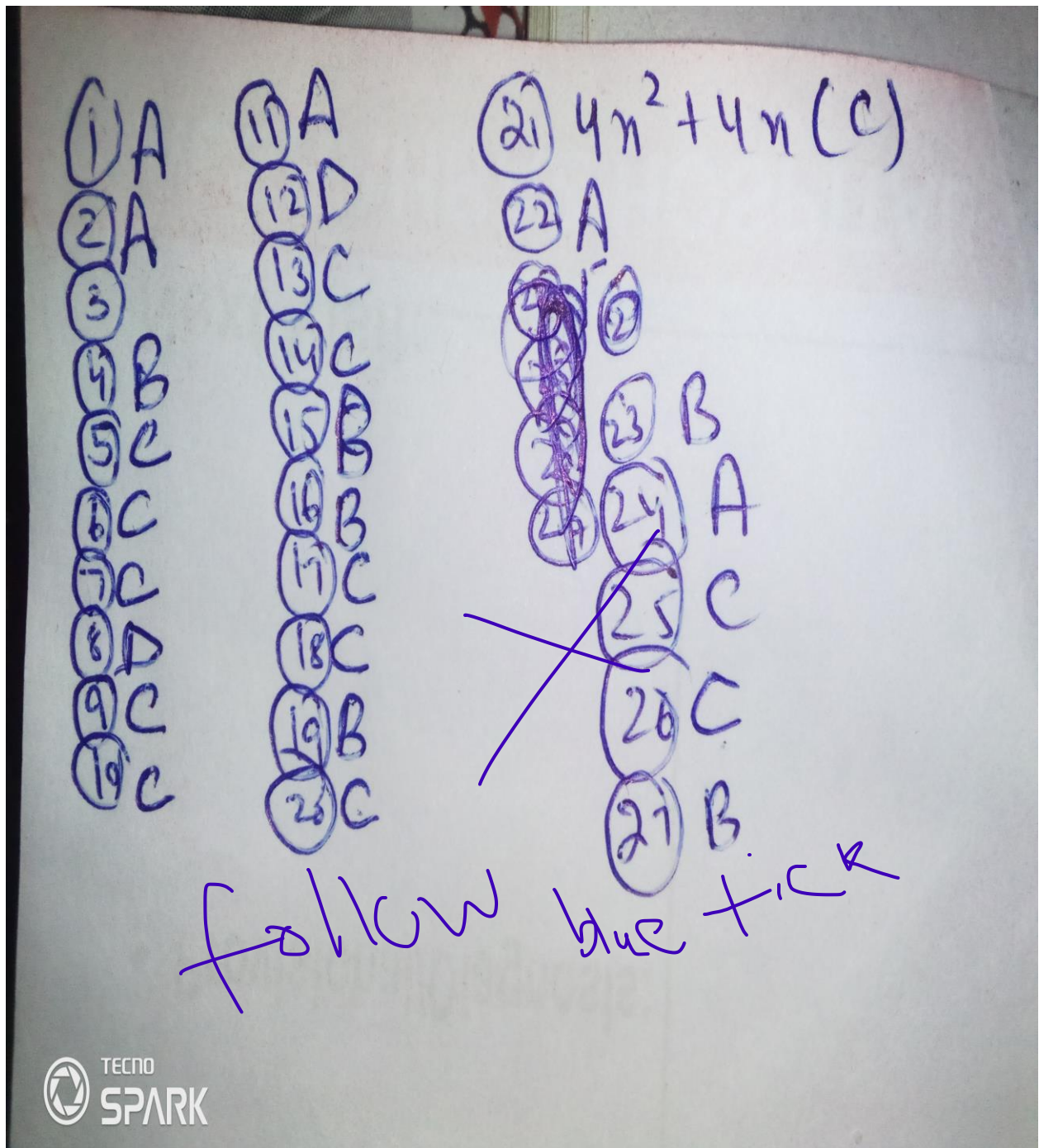


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Answer



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Q31

$$A = \{1, 3\} \quad B = \{1, 2, 3\}$$

A) order pair = ?

$$A \times B = \{(1, 1)(1, 2)(1, 3)(3, 1)(3, 2)(3, 3)\}$$

(b) Find domain and Range of R .

$$\text{let } R = \{(1, 1)(1, 2)(1, 3)(3, 1)(3, 2)(3, 3)\}$$

$$\text{Domain} = \{1, 3\} \quad \text{Range} = \{1, 2, 3\}$$

(c) check

$$(1, 2) \in R^2 \quad (3, 3) \in R^3$$

$$(1, 2) \in R^2 \text{ s.t. } \exists z \text{ such that } (1, 2) \in R \text{ s.t. } (2, 2) \in R.$$

$$(1, 2) \in R^2 \text{ becoz } (1, 2) \in R \text{ s.t. } (2, 2) \notin R$$

$$(3, 3) \text{ in } R^3 \text{ if } \exists z \text{ } (3, 2) \text{ belong to } R^2.$$

$$\text{and } (2, 3) \text{ belong to } R. \text{ in}$$

$$\text{case } (3, 3) \text{ is } R^3 \text{ becoz } (3, 1) \text{ belong to } R \text{ and } (1, 3) \text{ belong to } R.$$

$$\text{so yes } 1R_1 \text{ s.t. } 3R_3$$

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Q_{30}

$$A = \{2, 4, 6, 8, \dots, 18\} \quad B = \{3, 6, 9, 12, 15, \dots, 27\}$$

$$\text{L.H.S} = A \cap B = B \cap A$$

$$\text{L.H.S} = A \cap B$$
$$A \cap B = \{2, 4, 6, \dots, 18\} \cap \{3, 6, 9, 12, \dots, 27\}$$
$$A \cap B = \{6, 12, 18\}$$

$$\text{R.H.S} = B \cap A$$

$$B \cap A = \{3, 6, \dots, 27\} \cap \{2, 4, \dots, 18\}$$

$$B \cap A = \{6, 12, 18\}$$

$$\text{L.H.S} = \text{R.H.S} \quad \text{Proved}$$

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$$a_n = a_1 + (n-1)d$$

$$Q_{23} \quad a_n = a_1 + (n-1)d$$

$$a_9 = ?$$

$$a_1 = 10$$

$$n = 9$$

$$d = a_2 - a_1 = 15 - 10 = 5$$

$$a_9 = a_1 + (n-1)d$$

$$a_9 = 10 + (8)5$$

$$a_9 = 10 + 40$$

$$a_9 = 50$$

man deposit 50 in 9th month.

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Q 32

$$f(x) = 3x^9 + 7$$

$$y = 3x^9 + 7$$

Convert x into y

$$x = 3y^9 + 7$$

$$x - 7 = 3y^9$$

$$\frac{x-7}{3} = y^9$$

$$y = \left(\frac{x-7}{3} \right)^{\frac{1}{9}}$$

$$f^{-1}(x) = y$$

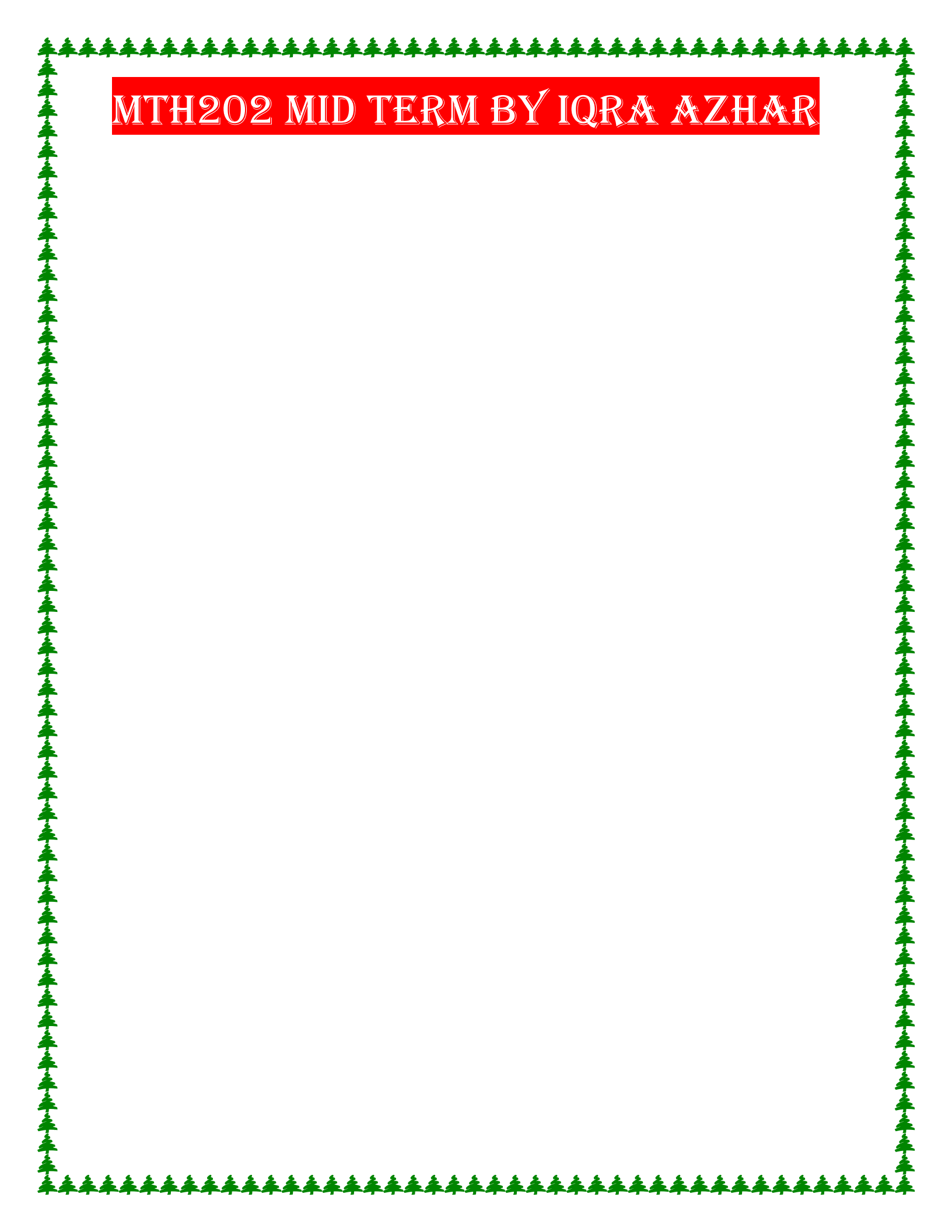
$$f^{-1}(x) = \left(\frac{x-7}{3} \right)^{\frac{1}{9}}$$

$$f^{-1}(-1) = \left(\frac{-1-7}{3} \right)^{\frac{1}{9}}$$

$$= \left(\frac{-8}{3} \right)^{\frac{1}{9}}$$

$$f^{-1}(-1) = \frac{-134217728}{19683}$$

$$= -6818.9$$



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